

Conjugacy of local homeomorphisms on totally disconnected spaces

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Project Outline

Dynamical systems $\rightarrow$ operator algebras

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Operator algebraic results $\rightarrow$ abstract algebraic results

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Progress made by Armstrong, Brix, Carlsen, Eilers in the C^* -algebraic context [ABCE23].

Deaconu–Renault Systems

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Assumption: X is totally disconnected (e.g. $\mathbb{Q}$, the Cantor set)

Write $\sigma_X^n := \underbrace{\sigma_X \circ \dots \circ \sigma_X}_n$.

Groupoids

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Range and domain maps $r, d: \mathcal{G} \rightarrow \mathcal{G}^{(0)}$.

Groupoid elements $\alpha, \beta \in \mathcal{G}$ composable if $r(\beta) = d(\alpha)$, in which case we denote their product by $\alpha\beta$.

A commutative diagram illustrating the composition of two morphisms α and β in a groupoid. The diagram shows three objects x , y , and z arranged horizontally. A horizontal arrow labeled α points from y to x . A horizontal arrow labeled β points from z to y . A curved arrow labeled $\alpha\beta$ points directly from z to x , representing the composition of β followed by α .

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If $\mathcal{G}$ admits a basis of compact open bisections, then we say $\mathcal{G}$ is ample, in which case $\mathcal{G}^{(0)}$ is necessarily totally disconnected.

Graded Groupoids

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A homomorphism of graded groupoids is a groupoid homomorphism $\psi: (\mathcal{G}_1, c_1) \rightarrow (\mathcal{G}_2, c_2)$ such that the diagram below commutes.

$$\begin{array}{ccc} \mathcal{G}_1 & \xrightarrow{\psi} & \mathcal{G}_2 \\ & \searrow c_1 \quad \swarrow c_2 & \\ & \Gamma & \end{array}$$

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Equivalently – we can write $\mathcal{G} = \sqcup_{\gamma \in \Gamma} \mathcal{G}_\gamma^c$, where $\mathcal{G}_\gamma^c := c^{-1}(\gamma)$ is the **homogeneous component of degree γ** .

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The **Steinberg algebra** (or **groupoid algebra**) of $\mathcal{G}$ is defined as

$$A_R(\mathcal{G}) := \text{span}\{1_B : B \text{ a compact open bisection of } \mathcal{G}\},$$

equipped with a convolution $*$ such that if U, V are compact open bisections of $\mathcal{G}$, then we set $1_U * 1_V = 1_{UV}$, where $UV = U_d \times_r V$.

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Extend linearly to $A_R(\mathcal{G})$, under which $A_R(\mathcal{G})$ is an R -algebra.

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The **canonical cocycle** on $\mathcal{G}_X$ is given by

$$c_X(x, m - n, y) := m - n,$$

which gives $\mathcal{G}_X$ the structure of a $\mathbb{Z}$ -graded groupoid.

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$$A = \bigoplus_{\gamma \in \Gamma} A_{\gamma},$$

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Example

Let $\mathcal{P}(R)$ be the ring of polynomials over a commutative ring R . Then $\mathcal{P}(R)$ is a $\mathbb{Z}$ -graded algebra, with $\mathcal{P}(R)_n = \{ax^n : a \in R\}$.

Graded Groupoid Algebras

If $(\mathcal{G}, c)$ is a Γ -graded groupoid, then $A_R(\mathcal{G})$ is automatically Γ -graded, by setting

$$A_R(\mathcal{G})_\gamma^c := \text{span}\{1_B : B \subseteq c^{-1}(\gamma) \text{ is a compact open bisection}\},$$

and it follows that

$$A_R(\mathcal{G}) = \bigoplus_{\gamma \in \Gamma} A_R(\mathcal{G})_\gamma^c.$$

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Similarly, when our groupoid algebras are graded, we are often interested in the case where maps preserve these gradings.

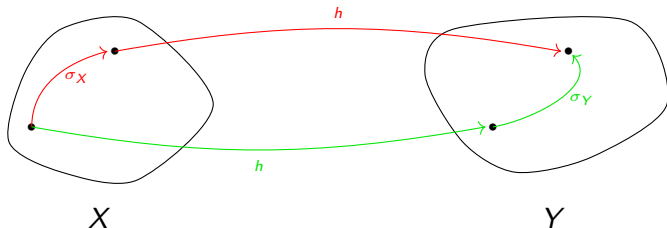
Conjugacy of Deaconu–Renault Systems

We say that two systems (X, σ_X) and (Y, σ_Y) are **conjugate** if there exists a homeomorphism $h: X \rightarrow Y$ such that

$$h \circ \sigma_X = \sigma_Y \circ h.$$

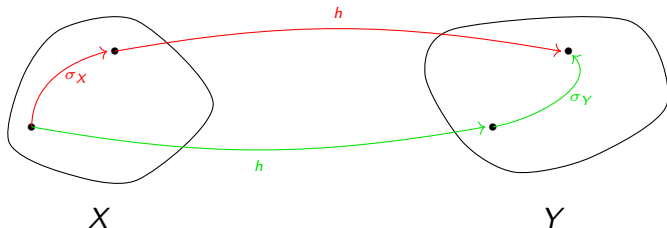
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Is it possible to take two Deaconu–Renault groupoids or their algebras, and decide whether the underlying systems are conjugate?

Weighted Cocycles

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Algebraic	Analytic
X lch + t.d	X lch
locally constant $f: X \rightarrow \Gamma$	continuous $f: X \rightarrow \Gamma$
$\mathcal{G}_X$ ample	$\mathcal{G}_X$ étale
$A_R(\mathcal{G}_X)$ an R -algebra	$C^*(\mathcal{G}_X)$ a C^* -algebra
Gradings	Dual actions

A collection $\mathcal{B}$ of open subsets of a space X is **separating for X** if, given $x \neq y \in X$, there exists $U \in \mathcal{B}$ such that $x \in U$ and $y \notin U$.

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Theorem

Let (X, σ_X) and (Y, σ_Y) be **totally disconnected** Deaconu–Renault systems. Let $\mathcal{B}$ be a collection of compact open subsets of X that is separating for X . Then, the following are equivalent.

1. The systems (X, σ_X) and (Y, σ_Y) are conjugate.
2. There exists a groupoid isomorphism $\psi: \mathcal{G}_X \rightarrow \mathcal{G}_Y$ satisfying the following equivalent conditions.
 - 2.1 $c_X = c_Y \circ \psi$, and ψ restricts to a conjugacy between X and
 - 2.2 There is a homeomorphism $h: X \rightarrow Y$ such that $c_{1_U \circ h} = c_{1_U} \circ \psi$ for all $U \in \mathcal{B}$.

Theorem

Let (X, σ_X) and (Y, σ_Y) be totally disconnected Deaconu–Renault systems, and $\mathcal{B}$ a collection of compact open subsets that is separating for Y . Then, the following are equivalent.

1. there is a conjugacy $h: X \rightarrow Y$;
2. there is an R -algebra isomorphism $\varphi: A_R(\mathcal{G}_X) \rightarrow A_R(\mathcal{G}_Y)$ and a homeomorphism $h: X \rightarrow Y$ such that

$$\varphi(A_R(\mathcal{G}_X)_{\gamma}^{c_{1_U \circ h}}) = A_R(\mathcal{G}_Y)_{\gamma}^{c_{1_U}}$$

for all $U \in \mathcal{B}$ and $\gamma \in \mathbb{Z}$.

Further Reading

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5. *Diagonal-preserving graded isomorphisms of Steinberg algebras* [CR18]



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